UNIVERSITAT POLITÈCNICA DE CATALUNYA BARCELONATECH

GIA - IAA
Introducció a l'Aprenentatge Automàtic

3. Lab: Dimensionality Reduction

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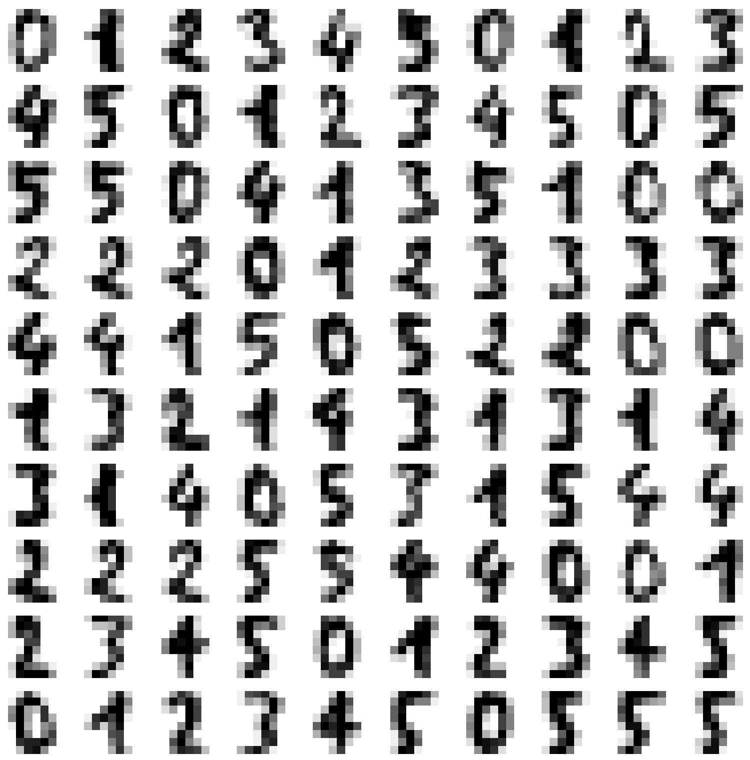
MNIST dataset

- We will use a reduced version of MNIST dataset (Modified National Institute of Standards and Technology database) is a large database of handwritten digits that is commonly used for training various image processing system

A selection from the 64-dimensional digits dataset



https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_digits.html

MNIST dataset

```python
from sklearn.datasets import load_digits

digits = load_digits(n_class=3)
X, y = digits.data, digits.target
n_samples, n_features = X.shape
n_neighbors = 30
import matplotlib.pyplot as plt
plt.gray()
plt.matshow(digits.images[0])
plt.show()
X.shape

```

✓ 0.1s

```
<Figure size 640x480 with 0 Axes>
```

A selection from the 64-dimensional digits dataset

API Reference > sklearn.datasets > load_digits

load_digits

`sklearn.datasets.load_digits(*, n_class=10, return_X_y=False, as_frame=False)` [source]

Load and return the digits dataset (classification).

Each datapoint is a 8x8 image of a digit.

Classes	10
Samples per class	~180
Samples total	1797
Dimensionality	64
Features	integers 0-16

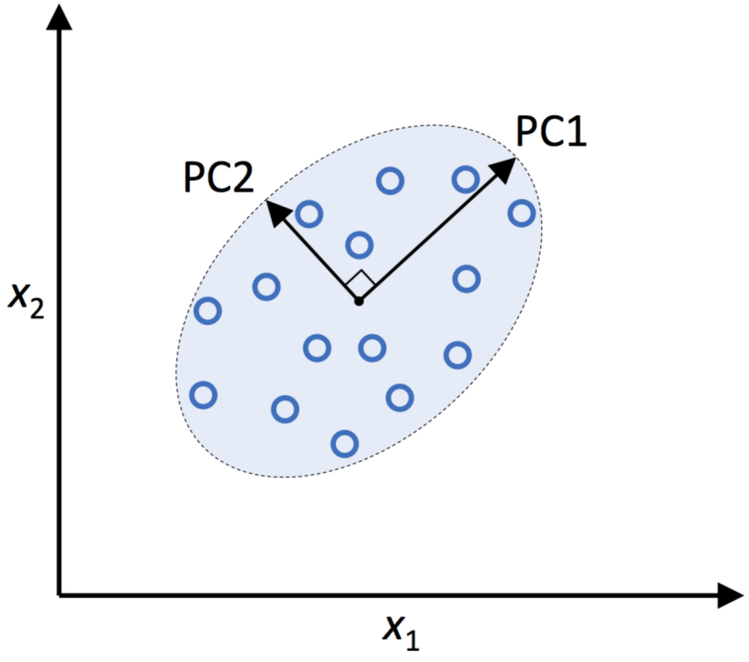
This is a copy of the test set of the UCI ML hand-written digits datasets https://archive.ics.uci.edu/ml/datasets/Optical+Recognition+of+Handwritten+Digits

Read more in the User Guide.

https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_digits.html

PCA in a nutshell

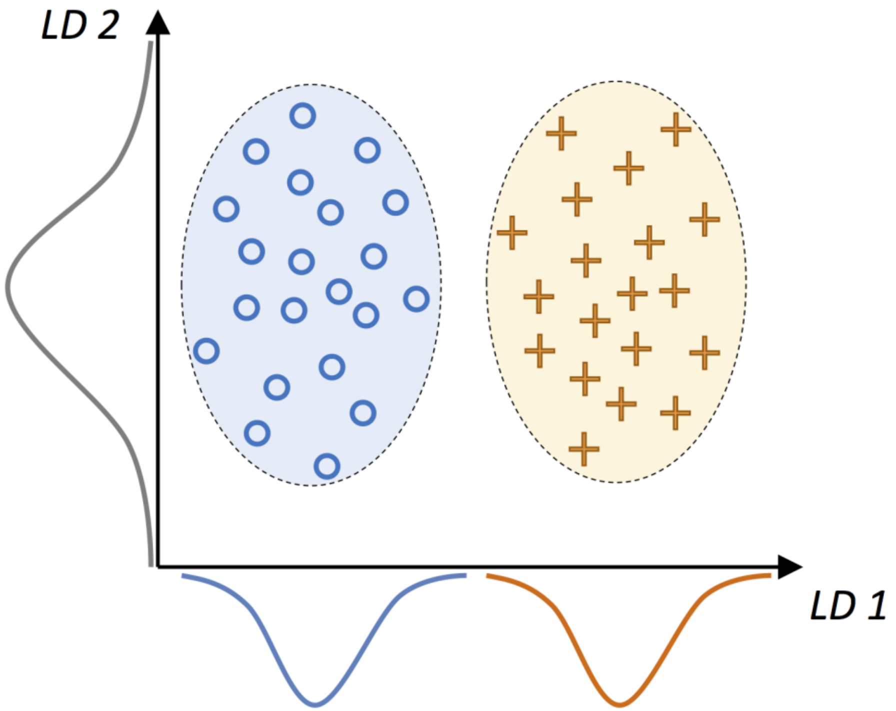
- PCA help us <u>identify</u> patterns in data based on the correlation between features
- PCA aims to find directions of maximum variability in high dimensional space and projects the data onto a new subspace with equal of fewer dimensions
- The orthogonal axes (PCA components) of the new subspace can be interpreted as the directions of maximum variance in the original data space



Chapter 5. Bishop, Christopher M. Pattern recognition and machine learning.

LDA in a nutshell

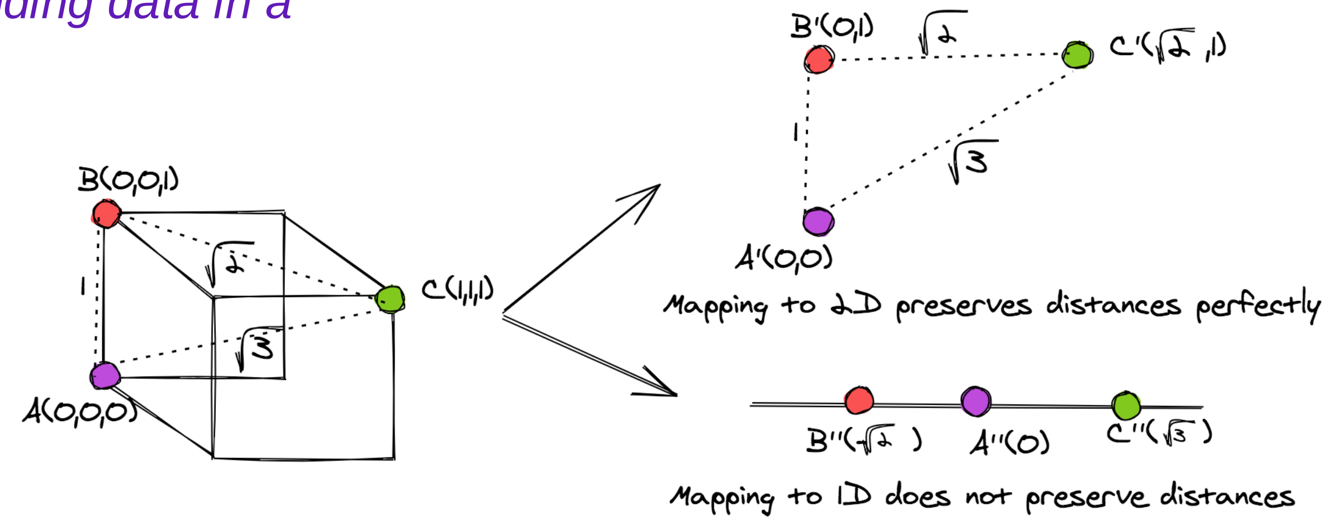
- Linear Discriminant Analysis operates by projecting data from a multidimensional graph onto a linear graph
- The easiest way to conceive of this is with a graph filled up with data points of two different classes
- When LDA is carried out there are two primary goals: minimizing the variance of the two classes and maximizing the distance between the means of the two data classes
- LDA works best when the means of the classes are far from each other. If the means of the distribution are shared it won't be possible for LDA to separate the classes with a new linear axis



Chapter 5. Bishop, Christopher M. Pattern recognition and machine learning.

MDS in a nutshell

MDS is a non-linear technique for embedding data in a lower-dimensional space..



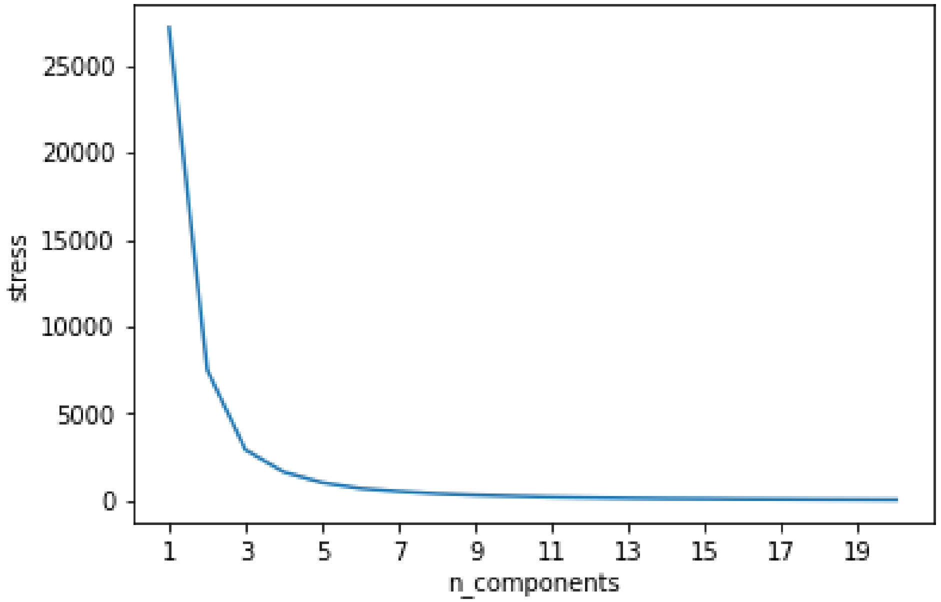
A possible mapping of points from 3D to 2D and 1D. The mapping is not optimized.

❖ It maps points residing in a higher-dimensional space to a lower-dimensional space while preserving the distances between those points as much as possible. Because of this, the pairwise distances between points in the lower-dimensional space are matched closely to their actual distances

https://scikit-learn.org/stable/modules/generated/sklearn.manifold.MDS.html#sklearn.manifold.MDS

MDS in a nutshell

- One of the important hyper-parameters involved in MDS is the size of the lower-dimensional space in which the points are embedded.
- This would be very relevant when MDS is used as a pre-processing step for dimensionality reduction.
- Just how many dimensions without losing important information?
- A **simple method to choose a value of this parameter** is to run MDS on different values of n_components and plot the stress_ value for each embedding. Given that the stress_ value decreases with higher dimensions - you pick a point that has a fair tradeoff between stress_ and n_components.



t-SNE for dimensionality reduction

- t-SNE is a popular dimensionality reduction algorithm that arises from probability theory. Simply put, it projects the high-dimensional data points (sometimes with hundreds of features) into 2D/3D by inducing the projected data to have a similar distribution as the original data points by minimizing something called the KL divergence
- Compared to a method like Principal Component Analysis (PCA), it takes significantly more time to converge, but present significantly better insights when visualized
- The main purpose of t-SNE is visualization of high-dimensional data. Hence, it works best when the data will be embedded on two or three dimensions

t-SNE: What is perplexity?

- t-SNE Perplexity is a measure for information that is defined as 2 to the power of the Shannon entropy. In t-SNE, the perplexity may be viewed as a knob that sets the number of effective nearest neighbors. Thus, it is comparable with the number of nearest neighbors k that is employed in many manifold learners.
- Optimizing the KL divergence can be a little bit tricky sometimes. There are five parameters that control the optimization of t-SNE and therefore possibly the quality of the resulting embedding:
    - perplexity
    - early exaggeration factor
    - learning rate
    - maximum number of iterations

https://scikit-learn.org/stable/modules/manifold.html#t-sne

UMAP and n_neighbors

- UMAP (Uniform Manifold Aproximation and Projection) is a fairly flexible non-linear dimension reduction algorithm. It seeks to learn the manifold structure of your data and find a low dimensional embedding that preserves the essential topological structure of that manifold.

- UMAP has several hyperparameters that can have a significant impact on the resulting embedding:
  - n_neighbors
  - min_dist
  - n_components
  - metric

https://umap-learn.readthedocs.io/en/latest/parameters.html